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SLO 5 - Team Work

Team Roles and Leadership

Each team member was given a chance to be the team lead both semesters. A team lead's sprint would end on Monday and the torch would be passed on to the next person. Any stories completed by Monday were part of last week's sprint. The most important task for the team lead was to make sure that there was a clear goal and that everyone understood the goal for that week and was motivated to work towards that. Failure to do this would leave us feeling less productive when the week was finished.

There was a time in the semester when we were too focused on training and testing the model, but then Logan was made team lead and he gave us a better goal. Part of his goal was to add and fix some other features, but he also just wanted us to have a model on the app, even if it wasn't good. This gave us all a common goal that we had agreed on and our productivity was up that week.

Stories were assigned based on the goal for that week's sprint. In the middle of the semester, one of our goals was to refactor and improve the screener. This didn't just mean the looks of it, but also its utility. While it was the team lead's role to give everyone a goal and hand out stories, it was on everyone to propose and groom new stories.

Story proposals gave other team members a say in what our goals should be for sprints and epics. Failure to propose and create stories would put everything on the team lead leaving them overwhelmed leading to less productivity. During the start of the semester, Nathan was the assigned team lead and he required that we propose/groom

a story before each meeting. This helped us stay focused and put less pressure on the team lead. This is a practice that we have continued doing since that sprint.

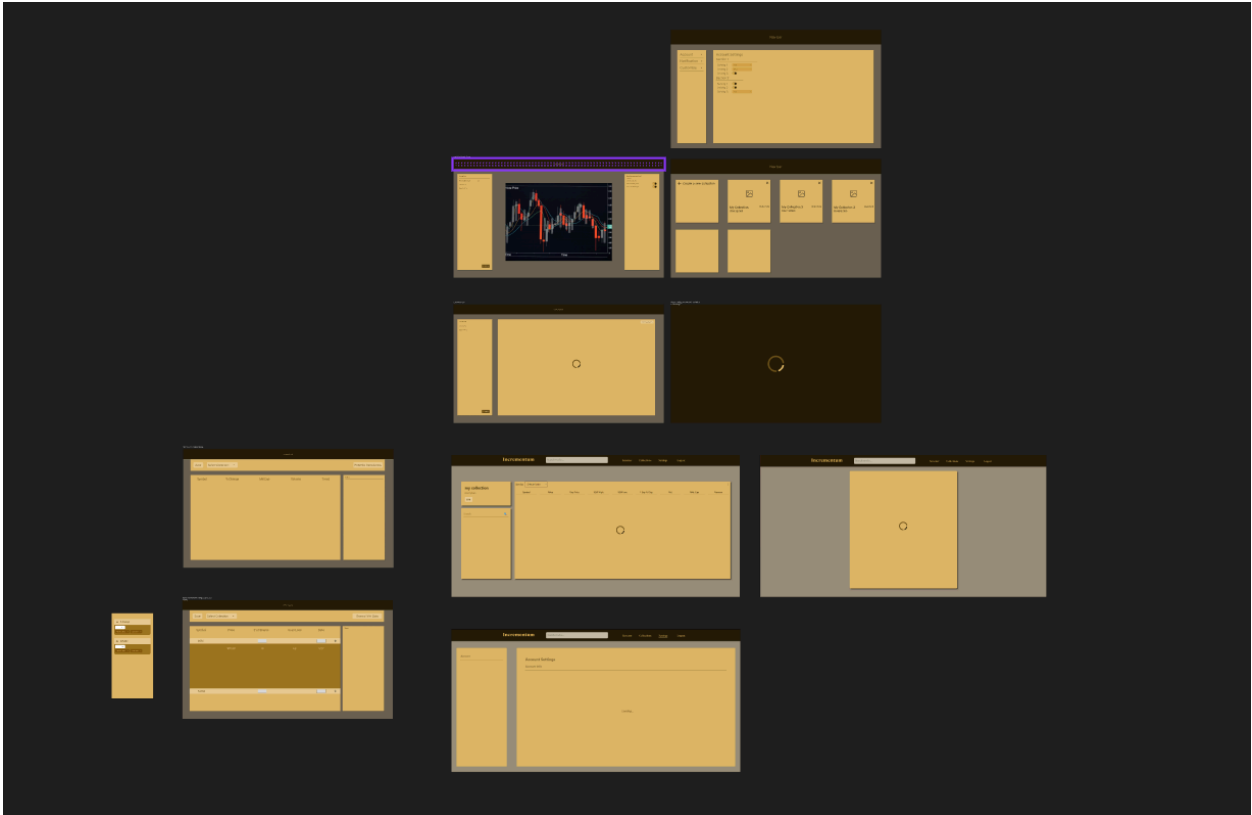
Team members could also give recommendations on stories that needed to be completed to help the team lead make decisions. There was one time that Logan wasn't sure what stories to assign and Chris recommended some stories because he thought that it would help us accomplish our goals, which kept us motivated for the rest of the week.

Collaborative and Inclusive Environment

Our team was very good at asking for the opinions of the other team members. Everyone was able to voice their opinion about our design and we all made design choices as a team. Each team member also had the opportunity to make a design and present it to the group.

This gave everyone involvement in the design decisions that we made for our app, and we were able to improve the app a lot because of it. At the beginning of this school year, we weren't as good about involving everyone, which is why our design wasn't great at first. We realized that we needed to have these disagreements and debates because they would give us the drive that we needed to improve our app.

Oftentimes, we did disagree on design choices. Sometimes we weren't able to completely settle debates, so the easiest way to settle it was to put it into practice. If not everyone agreed and we couldn't settle on something, we would go with the best option and experiment with it. User testing would give us an idea of how our design works and so it was easy to see errors in our designs.





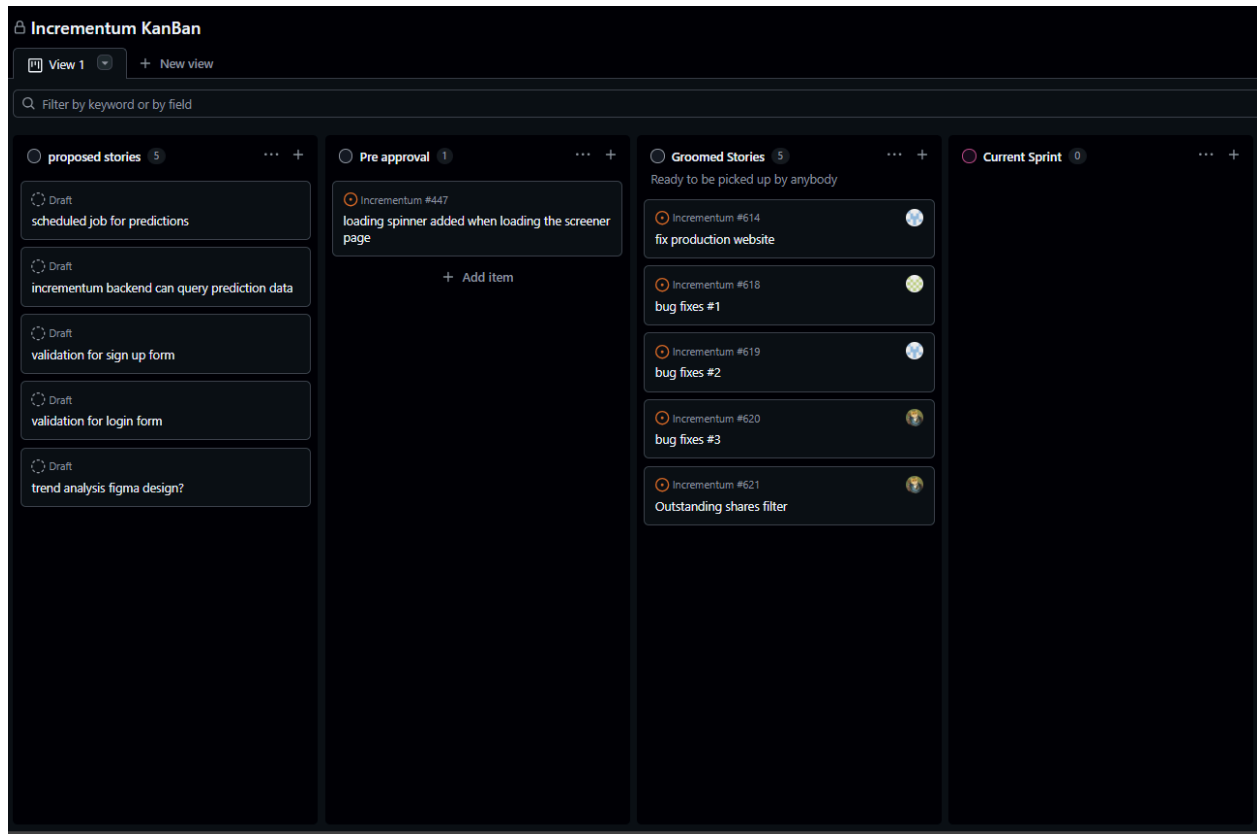
Goal Setting and Task Planning

We came up with a lot of the requirements for our app, but we would often look to our instructor for guidance as a product owner. He was able to take a look at our app and give us feedback on what we could improve on or add for future Epics. When starting Epics, we would come up with more detailed plans and write out our list of requirements. We kept a requirements spreadsheet and made sure to cross items off as we completed them to boost team morale.

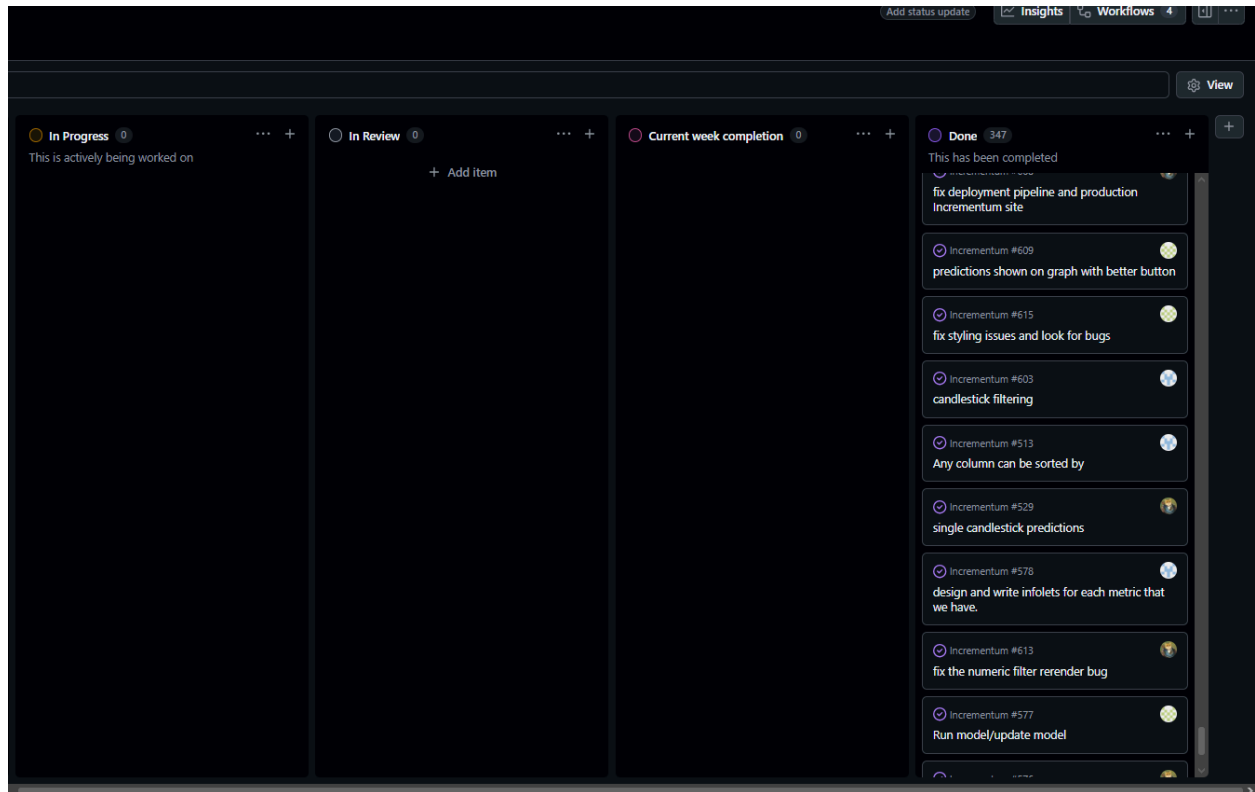
We would also create planning sheets for every sprint up to that Epic and would keep track of how closely we had been following those sprint plans. For epic 1, we were very unorganized and weren't sure at first if we could have all these features done by the time we were supposed to have them passed off, so we made a daily progress tracker to help keep us on track before we hit the big deadline.

This is something that we would try again because it worked well for us. Especially if we ended up being a part of the same team again, because we know what motivates us and keeps us moving. Please see the attached daily progress tracking sheet and the requirements doc spreadsheet.

For this project, we used a github project board and would assign tasks using it. We would discuss what each person's task was in the meeting, but we also had it up in this github project so everyone could look back at the acceptance criteria for their story. We would propose stories in one column, submit them for approval in the next column, and put the stories in groomed when they were approved. Team leads would then throw stories from groomed into the Current Week Sprint column and hand out stories from that column.



When stories were assigned, we would assign them the stories using the github project. We moved stories to the in progress column when we were actively working on them. When we felt like our story was finished, we would move them to the In Review column and have the pull request up and ready for the next meeting. If the story was completed, we would add it to our current week completion to record how well the week went. When it was a new week, these stories would be thrown into the Done column and it would be the next team lead's turn to keep track of the completed stories.



At the beginning of the semester, we still weren't very good at keeping our meetings short, but we improved a lot over this semester. We found a rhythm for each meeting and we spent a lot less time demoing new features to each other. Our meetings would be no more than 30 minutes most, if not all, of the time.

Every meeting would start with a review of the stories we each finished. We would quickly describe what we did and team members would quickly look over the pull request. Once we reviewed all of the stories, the team lead would then assign stories and then give the other team members time to ask questions. The team lead would also be recording the metrics we had for dividing merit money in the spreadsheet.

Last semester, we had a merit money sheet for recording the weekly stats. The problem with this sheet was that it relied on people actually keeping track of their hours

wasn't as effective.

[illegible]

week.

[illegible]

We tried to make it as simple to record as possible so we would actually take the time to record the metrics. We felt like the code quality metric wasn't the best metric for recording quality, but we only just realized that during one of our final sprints. One improvement that we could make is to split the code quality into two different metrics: clean and required test cases. Clean code would be a score and the required test cases would just be a boolean value to keep us accountable and ensure that we'd completed the required test cases. Both merit money spreadsheets have been attached with this report.

Finally, any leftover meeting time would be to discuss important items of business like plans for next Epics or suggestions for improving the design of our app. This time allowed each team member to voice their concerns and we were able to plan ahead for the sprints to come.

Meeting Objectives

We didn't accomplish everything that we set out to do. We aren't as confident in our prediction model and the features that we have in our app, but we got closer to having it ready. Although we haven't completed the predictions feature, we are happy with everything else that we created. Our idea for the app has changed a lot since the beginning of last semester and we managed to construct an intuitive app that has most, if not all, of the functionality that we wanted to add.

One of the biggest struggles with meeting our objectives was when we failed to communicate and we weren't all on the same page. To get ourselves back on track, we had received a lot of counsel from our instructor and we made sure that everyone was

involved in design decisions to boost team morale. Sometimes we were eager to code the fancy features that we were wanting to add, but our instructor would recommend alternatives like fixing the features we already had to support the new features.

We would often have our instructor put on a product owner hat and we would come up with a list of requirements together. We would plan smaller epics using these requirements and plan our weekly sprints around them. Our instructor gave us a better direction to go with our app, and while we didn't get to everything we wanted to accomplish, we were actually happy with the work we were doing.

This year was full of learning experiences, but we definitely learned the most from the experiences we had this semester. We had a lot of different influences on our apps design from user testing, our practicum instructor, and our machine learning instructor. We learned how to keep each other motivated and accountable for our work.

One of the most important lessons we learned was how important the design and research phases were. If we knew what we wanted from the start, then we wouldn't have encountered most of these issues that we had. Chris said that he would've spent more time researching api options and actually tested the different options (at least the ones that weren't blocked by a pay wall). We may have started using polygon a lot earlier in the project if we had.

Logan said that he would spend a lot more time in documentation when researching programming related tasks. If we had spent more time in the yfinance documentation, we would have realized earlier that yfinance wasn't the most reliable api for what we were doing. Finally, Nathan said that he would use wikipedia more for research and spend less time getting help from the AI overview. Wikipedia can help you

get a broad amount of information like documentation in the links and references. Our mistakes from this semester were valuable learning moments. The lessons we have learned apply to all software projects and will help us be more effective in projects to come.